SE_LAB05 : STATIC CODE ANALYSIS

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SECTION : 5_H

GIT REPOSITORY : https://github.com/yeasaswani-creator/SE_LAB_PES2UG24CS441_SEC_H

Pylint Report — Before Fixes

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat pylint_report_before.txt
************* Module shuttle_system
shuttle_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
shuttle_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:11:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
shuttle_system.py:13:16: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
shuttle_system.py:16:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:19:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:23:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:29:8: W0702: No exception type(s) specified (bare-except)
shuttle_system.py:33:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:33:0: C0103: Function name "assignDriver" doesn't conform to snake_case naming style (invalid-name)
shuttle_system.py:33:17: W0613: Unused argument 'route_id' (unused-argument)
shuttle_system.py:37:0: C0116: Missing function or method docstring (missing-function-docstring)
shuttle_system.py:38:4: W0603: Using the global statement (global-statement)
shuttle_system.py:39:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
shuttle_system.py:39:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
shuttle_system.py:42:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
shuttle_system.py:42:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
shuttle_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)

------------------------------------------------------------------
Your code has been rated at 6.25/10 (previous run: 6.25/10, +0.00)
```

Bandit BEFORE

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat bandit_report_before.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-10-14 06:56:43.493540+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: ./shuttle_system.py:29:8
28                      flagged.append(r)
29              except:
30                  pass
31          return flagged

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: ./shuttle_system.py:35:11
34          drivers = ["Prakash", "Leela", "Imran", "Tom"]
35          return random.choice(drivers)
36

--------------------------------------------------
```

```
Code scanned:
        Total lines of code: 50
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 2
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 2
Files skipped (0):
```

Flake8 BEFORE:

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat flake8_report_before.txt
shuttle_system.py:11:1: E302 expected 2 blank lines, found 1
shuttle_system.py:13:80: E501 line too long (82 > 79 characters)
shuttle_system.py:16:1: E302 expected 2 blank lines, found 1
shuttle_system.py:19:1: E302 expected 2 blank lines, found 1
shuttle_system.py:23:1: E302 expected 2 blank lines, found 1
shuttle_system.py:29:9: E722 do not use bare 'except'
shuttle_system.py:33:1: E302 expected 2 blank lines, found 1
shuttle_system.py:37:1: E302 expected 2 blank lines, found 1
shuttle_system.py:45:1: E302 expected 2 blank lines, found 1
shuttle_system.py:60:1: E305 expected 2 blank lines after class or function definition, found 1
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle %
```

Pylint — AFTER :

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % python -m pylint shuttle_system.py > pylint_report_after.txt 2>&1
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat pylint_report_after.txt

------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Bandit — AFTER

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat bandit_report_after.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.9
Run started:2026-10-14 07:30:05.179782+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 80
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Flake8 — AFTER

```
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % python -m flake8 shuttle_system.py > flake8_report_after.txt 2>&1
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle % cat flake8_report_after.txt
(.venv) (base) sangadalayeasasawani@sangadalas-MacBook-Air L5_07_campus_shuttle %
```

ff

Issue	Type	Line(s)	Description	Fix Approach
Mutable default argument	Bug	12	`logs=[]` is shared across function calls, potentially retaining previous log entries	Changed the default to `None` and initialized the list inside the function.
Missing docstrings	Code quality	1, 12, 18, 24, 30, 42, 50	Module and functions lacked descriptive documentation.	Added module and function docstrings.
Bare exception handling	Bug / Code quality	29	A bare `except` hides errors and silently ignores failures.	Removed the broad exception handler and explicitly checked trip data.
Insecure random selection warning	Security warning	35	Random.choice() triggered Bandit B311, which warns against using standard pseudo-	Replaced it with `secrets.choice()`.

			random generators for security-sensitive purposes	
Incorrect function naming	Code quality	33	`assignDriver` does not follow Python's snake_case naming convention.	Renamed it to `assign_driver` and updated its function call.
Unsafe file handling pattern	Code quality	39–42	Files were opened without context managers or an explicit encoding	Used `with open()` and specified UTF-8 encoding.
Incorrect blank-line spacing	Formatting	Multiple lines	Several functions did not have the required blank-line spacing.	Corrected function spacing according to Flake8 conventions.
Excessive line length	Formatting	13	A line exceeded Flake8's 79-character limit.	Wrapped the long expression across multiple lines.
Route lookup after deletion	Runtime bug	60	The program attempted to calculate R3's average after removing R3, causing a `KeyError`	Checked whether R3 exists before accessing it.